1 Quantitative scale control for tangent maps

Source: `PrimeTensor/Analysis/Control.lean`

What this file establishes

Every tangent map so far has been an arbitrary morphism of the carrier. `PrimeTensor/Analysis/Derivative.lean` observed that nothing required one to be continuous, and that nothing up to that point needed it. The chain rule does, and the module header says why the obvious repair is insufficient: ordinary continuity at the pivot is not enough, because the derivative measures a *relative* rate, so what is wanted is a quantitative statement — an input refinement gain that delivers a prescribed output refinement gain, uniformly in the ambient level. That property is ScaleControlled, and this file introduces it, verifies it for every tangent map the development has built, and proves the two consequences the chain rule will consume.

Getting there requires a fact outstanding since `PrimeTensor/Scale.lean`: that the levels are *nested*, so that closeness at a fine level implies closeness at a coarse one. That file recorded the statement as true and unproved; this one proves it, and Remark 1.5 explains why it becomes unavoidable exactly here.

1.1 The levels are nested

Lemma 1.1 (`MulRat.scaleWithin_succ_weaken`)**.** ScaleWithin($\mathsf{succ}\,\ell, a, b$) *implies* ScaleWithin($\ell, a, b$).

```
theorem scaleWithin_succ_weaken {level : Depth} {a b : MulRat}
    (h : ScaleWithin (.succ level) a b) :
    ScaleWithin level a b := by
  unfold ScaleWithin at h ⊢

  have hone :
      (1 : MulRat) * (1 : MulRat) < two := by
    rw [one_mul]
    exact one_lt_two
```

Proof. Write $q = \text{scalePow}(\text{ratio}(a, b), \ell)$, so that the forward hypothesis is $q \cdot q < \text{two}$. Also $1 \cdot 1 < \text{two}$, by cancelling a unit and citing one_lt_two. Now apply mul_lt_two_of_sq_lt_two of `PrimeTensor/Scale/Laws.lean` to the pair $(q, 1)$: both have square below the ruler, so their product $q \cdot 1$ does, and cancelling the unit gives $q < \text{two}$, which is the forward half of the conclusion. The backward half is the same argument on the reversed ratio. □

Remark 1.2 (Monotonicity, and the route taken to it). `PrimeTensor/Scale.lean` stated this as something the ladder ought to satisfy, gave the informal argument — if $r^{2^{k-1}} \geq 2$ then $r^{2^k} \geq 4 > 2$ — and noted that the argument needed a comparison fact not then available. `PrimeTensor/Scale/Laws.lean` supplied that fact and observed the statement had become a corollary of the composition law with the middle magnitude taken to be a itself.

The proof above takes neither route. It pairs the hypothesis not with a second hypothesis but with the trivial bound $1 \cdot 1 < \text{two}$, and lets the arithmetic core do the rest. The effect is the same and the shape is more direct: weakening a level is composing with the identity comparison, so it needs no new inequality at all — only the observation that the pivot is itself within the ruler, which was the very first lemma of `PrimeTensor/Scale.lean`.

Lemma 1.3 (`MulReal.scaleNear_succ_weaken` and `scaleNear_weaken`)**.** $\mathrm{ScaleNear}(\mathtt{succ}\,\ell, a, b)$ *implies* $\mathrm{ScaleNear}(\ell, a, b)$*; and more generally, if* $\mathrm{AtOrAfter}(c, f)$ *then* $\mathrm{ScaleNear}(f, a, b)$ *implies* $\mathrm{ScaleNear}(c, a, b)$*.*

```
theorem scaleNear_weaken {coarse fine : Depth} {a b : MulReal}
    (hcf : Depth.AtOrAfter coarse fine)
    (h : ScaleNear fine a b) :
    ScaleNear coarse a b := by
  induction hcf with
  | here =>
      exact h
  | later hcf ih =>
      exact ih (scaleNear_succ_weaken h)
```

Proof. The first transports Lemma 1.1 across the quotient, keeping the same witnessing streams and the same anchor and applying it termwise. The second iterates: induction on the proof that the fine level lies at or after the coarse one peels off one `succ` at a time, each peel being one application of the first. □

So the family $\{\mathrm{ScaleNear}(\ell, -, -)\}$ is decreasing along AtOrAfter — the levels really are a nested system of tolerances, and the order on Depth built in `PrimeTensor/Completion.lean` for indexing tails now also orders tolerances.

1.2 Advancing respects the order

Lemma 1.4 (`Depth.advance_atOrAfter`, `advance_left_join` and `advance_right_join`)**.** *If* $\mathrm{AtOrAfter}(a, b)$ *then* $\mathrm{AtOrAfter}(\mathrm{advance}(\ell, a), \mathrm{advance}(\ell, b))$*. Consequently* $\mathrm{advance}(\ell, \mathrm{join}(a, b))$ *lies at or after both* $\mathrm{advance}(\ell, a)$ *and* $\mathrm{advance}(\ell, b)$*.*

Proof. The first is induction on the given proof, each `later` becoming one `later` on the advanced pair, since advance recurses on its second argument by a single `succ`. The two corollaries apply it to the bounds on join from `PrimeTensor/Completion.lean`. □

Remark 1.5 (Why monotonicity becomes unavoidable in this file)*.* Six earlier proofs have combined two hypotheses by taking a join, and none of them needed the levels to be nested. The reason is that in all six the join was taken in the *anchor* coordinate: an index lying in the tail at $\mathrm{join}(c_1, c_2)$ lies in the tail at c_1 by transitivity of AtOrAfter, and transitivity is all that is required.

Theorem 1.10 takes a join in the *level* coordinate instead — of two source gains. The hypothesis then arrives at $\mathrm{advance}(\ell, \mathrm{join}(s_1, s_2))$, which is *finer* than the $\mathrm{advance}(\ell, s_1)$ that the first control hypothesis expects, and a finer nearness is not automatically a coarser one. Converting it is exactly Lemma 1.3, applied through Lemma 1.4. That is why the statement outstanding since `PrimeTensor/Scale.lean` is proved here and not earlier: this is the first place a join happens among tolerances rather than among stages.

1.3 Scale control

Definition 1.6 (`MulTangentMap.ScaleControlled`)**.** A tangent map D is *scale controlled* when

$$\forall\, t\; \exists\, s\; \forall\, \ell\; \forall\, h: \quad \mathrm{ScaleNear}(\mathrm{advance}(\ell, s),\, h,\, 1) \;\longrightarrow\; \mathrm{ScaleNear}(\mathrm{advance}(\ell, t),\, D(h),\, 1).$$

```
def ScaleControlled (D : MulTangentMap) : Prop :=
  ∀ targetGain : Depth,
    ∃ sourceGain : Depth,
      ∀ level : Depth, ∀ h : MulReal,
        MulReal.ScaleNear (Depth.advance level sourceGain) h 1 →
        MulReal.ScaleNear (Depth.advance level targetGain) (D h) 1
```

Design note 1.7 (What the quantifier order buys)*.* The source gain s is chosen *before* the ambient level ℓ, so one s must work at every level simultaneously. Had the order been $\forall t\, \forall \ell\, \exists s$, the modulus could depend on the ambient scale and the definition would be a level-by-level continuity statement. As written it is uniform, and that uniformity is what the docstring calls finite scale distortion.

In the logarithmic chart of `PrimeTensor/Analysis/Derivative.lean` the reading is the familiar one. Writing $T = 2^{-(|\ell|-1)}$ for the tolerance at level ℓ, the hypothesis is $|\log h| < 2^{-|s|}\, T$ and the conclusion is $|\log D(h)| < 2^{-|t|}\, T$, for every dyadic T. Since the induced map on logarithms is additive, that says it is bounded, with $2^{|s|-|t|}$ serving as a bound on its operator norm. So ScaleControlled is boundedness of the tangent map — exactly the hypothesis `PrimeTensor/Analysis/Derivative.lean` declined to impose, imposed now that something needs it.

Lemma 1.8 (`scaleControlled_identity` and `scaleControlled_trivial`)**.** identity *and* trivial *are scale controlled.*

Proof. For the identity take $s = t$ and return the hypothesis unchanged — no distortion at all. For the trivial map take $s = t$ as well; the value is the pivot, and scaleNear_refl places it near itself at every level, so the hypothesis is discarded. ∎

Theorem 1.9 (`ScaleControlled.inv`)**.** *If D is scale controlled then so is D^{-1}, with the same source gain.*

Proof. Take the source gain D supplies for the same target, apply D's control, and invert the conclusion with scaleNear_inv of `PrimeTensor/Analysis/Near/Laws.lean`; the pivot is fixed by inversion, so the right-hand side stays 1. ∎

Theorem 1.10 (`ScaleControlled.mul`)**.** *If D and E are scale controlled then so is $D \cdot E$.*

```
theorem ScaleControlled.mul {D E : MulTangentMap}
    (hD : ScaleControlled D)
    (hE : ScaleControlled E) :
    ScaleControlled (mul D E) := by
  intro targetGain

  obtain ⟨sourceD, hcontrolD⟩ := hD (.succ targetGain)
  obtain ⟨sourceE, hcontrolE⟩ := hE (.succ targetGain)

  let sourceGain := Depth.join sourceD sourceE
```

Proof. Given a target gain t, ask each factor for $\mathsf{succ}\, t$ — one deeper, because the product will spend a level — obtaining source gains s_D and s_E, and offer $\mathrm{join}(s_D, s_E)$ as the source gain.

Let h be near the pivot at $\mathrm{advance}(\ell, \mathrm{join}(s_D, s_E))$. By Lemma 1.4 that level is at or after both $\mathrm{advance}(\ell, s_D)$ and $\mathrm{advance}(\ell, s_E)$, so Lemma 1.3 weakens the hypothesis to each of them and both controls apply, putting $D(h)$ and $E(h)$ near the pivot at $\mathrm{advance}(\ell, \mathsf{succ}\, t)$. That level is $\mathsf{succ}\, \mathrm{advance}(\ell, t)$ by advance_succ, so scaleNear_mul combines the two at $\mathrm{advance}(\ell, t)$; the right-hand side is $1 \cdot 1$, which cancels. □

Corollary 1.11 (`scaleControlled_depthPow`). *Every positive-depth tangent power* $\mathrm{depthPow}(d)$ *is scale controlled.*

Proof. Structural recursion: at `one` it is the identity, and at $\mathsf{succ}\, d$ it is the product of the identity with the recursive case. □

Remark 1.12 (Controlled is a property of the family, not of tangent maps). No theorem here says that a tangent map is scale controlled; the property is verified one construction at a time. It could not be otherwise — a tangent map is any morphism of the carrier whatever, and `PrimeTensor/Analysis/Derivative.lean` deliberately imposed no regularity, so a wild one is not excluded by anything. What Lemma 1.8, Theorems 1.9 and 1.10 and Corollary 1.11 establish jointly is that the *closure* of $\{\mathrm{identity}, \mathrm{trivial}\}$ under product and inverse consists of controlled maps — which, by `PrimeTensor/Analysis/Examples.lean`, is every tangent map the development has so far produced as a derivative. The tangent ratio is not separately checked, although it is a product with an inverse and so is covered.

1.4 The two consequences

Theorem 1.13 (`ScaleControlled.maps_converges`). *A scale-controlled D sends pivot-convergent sequences to pivot-convergent sequences.*

Proof. Fix a level ℓ. Ask the control for the smallest target gain, `one`, obtaining a source gain s, and ask the convergence hypothesis for the level $\mathrm{advance}(\ell, s)$, obtaining an anchor. Beyond it, the control puts $D(s_n)$ near the pivot at $\mathrm{advance}(\ell, \mathsf{one})$, which is $\mathsf{succ}\, \ell$; weakening once by Lemma 1.3 gives level ℓ, as required. □

This is continuity at the pivot, in the sequential form of `PrimeTensor/Analysis/Limit.lean`. The implication runs one way only: control is defined uniformly in the level, and nothing here derives it back from the sequential statement. That gap is the content of the header's remark that ordinary continuity is not enough.

Theorem 1.14 (`negligible_map`). *If D is scale controlled and* error *is negligible relative to* base*, then $n \mapsto D(\mathrm{error}_n)$ is negligible relative to* base*.*

```
theorem negligible_map
    {D : MulTangentMap}
    (hD : D.ScaleControlled)
    {error base : MulReal.Seq}
    (herror : NegligibleRelative error base) :
    NegligibleRelative (fun n => D (error n)) base := by
  intro targetGain
```

```
obtain ⟨sourceGain, hcontrol⟩ := hD targetGain
obtain ⟨anchor, htail⟩ := herror sourceGain

refine ⟨anchor, ?_⟩
intro n hn level hbase
```

Proof. Given a target gain, take the source gain the control supplies and the anchor the negligibility supplies at that source gain. Beyond the anchor, whatever level the base achieves, the error is near the pivot at the source-advanced level, and the control converts that into the target-advanced level for $D(\mathrm{error}_n)$. The two definitions match term for term; the proof is their composition and nothing more. □

Remark 1.15 (Everything here is for the chain rule). Theorem 1.14 is the reason the file exists. A chain rule must push the error of the inner function through the outer tangent map and still call it negligible, and negligibility is a statement about levels, so the outer map has to move levels predictably. Nothing in this file states a chain rule, composes two functions, or defines composition of tangent maps; all of that is downstream. What is settled here is the hypothesis such a rule will carry.

Remark 1.16 (What is not proved). ScaleControlled is not shown closed under composition of tangent maps, composition itself not being defined. There is no converse to Theorem 1.13, and no example of a tangent map that fails to be controlled — so the notion is not shown to be a genuine restriction, only a sufficient one. Nothing relates control to HasMulDerivativeAt: a derivative is not required to be controlled, and a function is not shown to have a controlled derivative.

Summary of the correspondence

Lean declaration	Kind	Here
`MulRat.scaleWithin_succ_weaken`	theorem	Lem. 1.1
`MulReal.scaleNear_succ_weaken`	theorem	Lem. 1.3
`MulReal.scaleNear_weaken`	theorem	Lem. 1.3
`Depth.advance_atOrAfter`	theorem	Lem. 1.4
`Depth.advance_left_join`	theorem	Lem. 1.4
`Depth.advance_right_join`	theorem	Lem. 1.4
`MulTangentMap.ScaleControlled`	definition	Def. 1.6
`scaleControlled_identity`	theorem	Lem. 1.8
`scaleControlled_trivial`	theorem	Lem. 1.8
`ScaleControlled.inv`	theorem	Thm. 1.9
`ScaleControlled.mul`	theorem	Thm. 1.10
`scaleControlled_depthPow`	theorem	Cor. 1.11
`ScaleControlled.maps_converges`	theorem	Thm. 1.13
`MulDifferential.negligible_map`	theorem	Thm. 1.14

Every declaration of `PrimeTensor/Analysis/Control.lean` appears above. The file contains no `sorry`, no axiom, and no unproved named proposition. Design note 1.7 and Remarks 1.2, 1.5, 1.12, 1.15 and 1.16 are stated for the reader and are not declarations of the file; the boundedness reading of scale control is a gloss and is not formalised.